

ENGINEERING SPECIFICATION

ES	030467-0	3995
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TITLE 1610-3 FUNCTIONAL SPECIFICATION

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PRODUCT 1610-3

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1610-3 FUNCTIONAL SPECIFICATION

ORIGINATOR					APPROVAL				
EC NO	3995								
DATE									

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1. SCOPE

This specification defines the design requirements for the Shugart 1610-3 Rigid Disk Controller.

2. APPLICABLE DOCUMENTS

All relevant Engineering Specifications.

Small Computer Systems Interface (ANSI X3T9.2) Specification
ES600 Disk Drive Engineering Specification.
ES700 Disk Drive Engineering Specification.

3. GENERAL DESCRIPTION**3.1 General Description.**

The 1610-3 Controller is a microprocessor based rigid disk controller with three custom LSI chips (NMOS). The controller is capable of controlling two Winchester disk drives, ST506 interface compatible. The dimensions of the controller are in the "foot print" of the drives. The custom LSI circuitry is implemented in the Buffer controller chip, Disk controller chip, and Encode/Decode chip. The 1610-3 Controller is designed as a peripheral controller to be used with the low end of the SCSI industry standard interface as defined by the SCSI specification no. (ANSI X3T9.2).

3.2 Features.

- * Support subset of SCSI commands on Host Adapter systems without Arbitration.
- * Support two different disk drives of the same capacities.
- * Up to 1-Mbytes/second maximum bus data transfer rate.
- * 5-Mbits/second (ST506) Drive transfer rate.
- * 1-KByte Dual Port data buffer.
- * Asynchronous REQ/ACK Handshake for bus data transfer.
- * Single Ended drivers/receivers.

- * Removable bus terminations.
- * Implied Seek for data transfer operations.
- * 21-Bit logical block addressing.
- * Automatic Cylinder/Head switching.
- * Programmable sector interleaving.
- * Jumper selectable block sizes (256, 512 bytes).
- * Multiple block transfer.
- * Automatic Read retries.
- * Track level defect handling.
- * 32-bit ECC polynomial for data field. Allows 11-bit burst error detection and 8-bit burst error correction.
- * 16-bit CRC for ID field verification.
- * Jumper selectable PULSE-MODE Selection.
- * Power On Diagnostic.
- * 20,000 hours MTBF

4. DISK DRIVE INTERFACE.

The drive interface is ST506 compatible. The control signals to and from the disk drives are transmitted via a daisy-chain cable at connector J2. Data and clock signals are transmitted differentially between the controller and the drives through the connectors J3, J4, and J5. Figure 4.1 shows the cable configuration between the controller and drives.

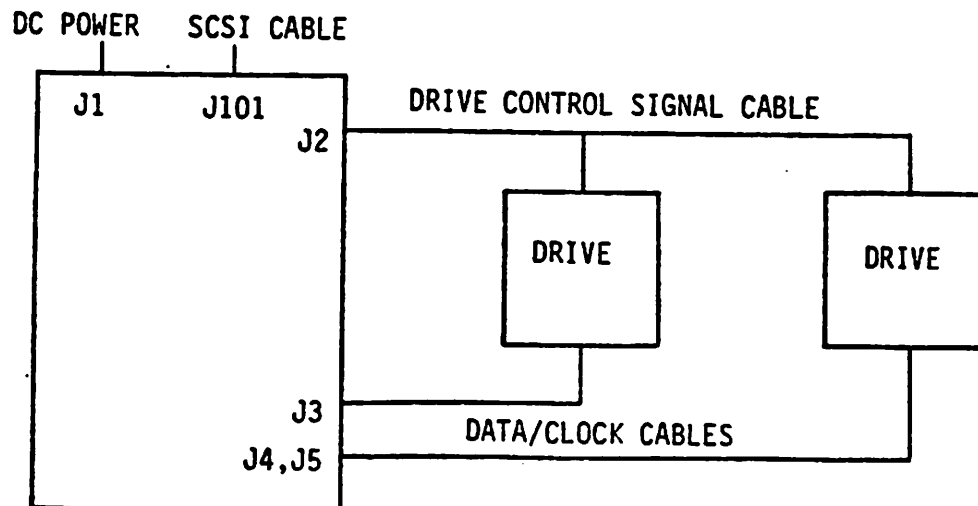


Figure 4.1 1610-3 Drive Cable Configuration.

4.1 Drive Control Signals.

J2 is a 34 pin, control signal, card edge connector. Maximum length should not exceed 20 feet (6 meters). Open collector

Note: All odd pins are connected to ground.

SIGNAL	PIN	DESCRIPTION
-RWC	2	-REDUCE WRITE CURRENT
-HS2	4	-HEAD SELECT BIT 2
-WG	6	-WRITE GATE
-SKC	8	-SEEK COMPLETE
-TK0	10	-TRACK ZERO
-FLT	12	-WRITE FAULT
-HS0	14	-HEAD SELECT BIT 0
-HS1	16	SPARE
-HS1	18	-HEAD SELECT BIT 1
-INDEX	20	-INDEX
-RDY	22	-DRIVE READY
-STP	24	-STOP
-DS1	26	-DRIVE SELECT 1
-DS2	28	-DRIVE SELECT 2
-DS3	30	-DRIVE SELECT 3
-DS4	32	-DRIVE SELECT 4
-DIR	34	-DIRECTION IN

Table 4.1. J2 Control Signal Cable Pin Assignments

4.1.1. drivers are used for all transmitted signals. All assigned control signals are terminated with 220 ohms to +5 volts and 330 ohms to ground at the receiving end of the cable. These termination resistor to be provided by a removable resistor pack. The pin assignment of the J2 connector is shown in Table 4.1.

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4.2 Data and Clock Signals.

J3 and J4 are 20 pin data/clock header pin connectors. J5 is the image of J4 in the form of edge connector. Maximum cable length should not exceed 20 feet (6 meters). Each connector serves one drive. Table 4.2 describes the pin signals of J3 ,J4 and J5.

All assigned data/clock lines are terminated with 100 ohms resistors from each side of the differential line to ground at the receiving end only.

Table 4.2. Data/Clock Signal Cable Pin Assignments.

SIGNAL	PIN	DESCRIPTION
-SEL	1	-DRIVE SELECTED
	2	GROUND
	3	SPARE
	4	GROUND
	5	SPARE
	6	GROUND
	7	SPARE
	8	GROUND
	9	SPARE
	10	SPARE
	11	GROUND
	12	GROUND
+WD	13	+ MFM WRITE DATA
-WD	14	- MFM WRITE DATA
	15	GROUND
	16	GROUND
+RD	17	+ MFM READ DATA
-RD	18	- MFM READ DATA
	19	GROUND
	20	GROUND

4.3 Cable Termination.

The last physical drive at the end of the J2 cable must be terminated.

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5. HOST BUS.

The 1610-3 controller communicates with host systems through an interface bus derived from the Small Computer Systems Interface (SCSI). The controller is connected to the host bus via the 50 pin connector at J101. This cable length should not exceed 20 feet (6 meters) from the first to the last device on the bus.

5.1 Interface Signals.

Refer to SCSI Specification (ANSI X3T9.2) for detail description of bus signals.

Table 5.1 shows the pin description of the J101 connector.

Table 5.1. SCSI/SASI Bus Cable Pin Description.

SIGNAL	PIN	DESCRIPTION
-DB0	2	-DATA BUS BIT 0
-DB1	4	-DATA BUS BIT 1
-DB2	6	-DATA BUS BIT 2
-DB3	8	-DATA BUS BIT 3
-DB4	10	-DATA BUS BIT 4
-DB5	12	-DATA BUS BIT 5
-DB6	14	-DATA BUS BIT 6
-DB7	16	-DATA BUS BIT 7
-DP	18	-PARITY, (NOT USED,TERMINATED)
	20	GROUND
	22	GROUND
	24	GROUND
	26	OPEN
	28	GROUND
	30	GROUND
-ATN	32	-ATTENTION
-GND	34	-GROUND
-BSY	36	-BUSY
-ACK	38	-ACKNOWLEDGE
-RST	40	-RESET
-MSG	42	-MESSAGE
-SEL	44	-SELECT
-C/D	46	-CONTROL/DATA
-REQ	48	-REQUEST
-I/O	50	-INPUT/OUTPUT

Note: All odd pins except pin 25 (open) are connected to ground.

5.2 Interface Protocol.

For detailed information regarding the Host Interface Protocol, refer to the SCSI Specification (ANSI X3T9.2)

5.3 Cable Termination.

All assigned signal lines are terminated with 220/330 ohm resistor packs at the controller side. These resistor packs are mounted on sockets to allow for removal if controller is used in the middle of the bus cable.

5.4 Bus Device ID.

The controller is allowed to assume any uniquely assigned SCSI bus device address. This device address (from 0 to 7) can be selected through the jumpers CU4, CU2, CU1. Refer to diagram 6.1 for the location of these jumpers.

6. CIRCUIT BOARD.

The 1610-3 controller consists of a single 2-layer printed circuit board. Figure 6.1 shows the locations of the various cable connectors, removable terminators and option jumpers.

6.1 Option Jumpers.

6.1.1 CU4, CU2, CU1: This group of jumpers are used to set device address of the controller on the SCSI bus.

6.1.2 Jumper W: This jumper is used to select the size of the sectors.

J-W	Bytes/Sec	Sectors/Trk
Out	256	32
✓ In	512	17

6.1.3 Jumpers X and Y: These jumpers are not used.

6.1.4 Jumper Z: This jumper is used in conjunction with the Selection mode jumpers SX, SH and PS. This jumper should be installed if the PULSE MODE is selected. See sections 6.1.7 and 6.1.8.

6.1.5 Jumper H2-H1: These positions if jumpered select

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the Hardware Reset mode. When a SCSI host device issues a RESET on the bus, the microprocessor of the controller will be reset. This has the same effect as a power-on reset.

- 6.1.6 Jumper H2-S: These positions if jumpered select the Software Reset mode. The RESET signal from the host device is latched by the controller. The microprocessor, under program control, would perform a controller reset on the SCSI bus.
- 6.1.7 Jumper SX-NS: These positions if jumpered select the normal mode of SCSI bus device selection. During the selection phase, the host must continue to assert the SELECT signal until the controller responds with the BUSY signal. If this mode is selected, jumper Z must not be installed.
- 6.1.8 Jumper SX-PS: This jumper position allows the host adapter to select the controller by pulsing the SELECT line (PULSE MODE). The SELECT signal is latched by the controller allowing the microprocessor to detect that the host initiated selection. This enables the controller to respond to a host that issues short selection pulses. Jumper Z must be installed if this mode is selected.

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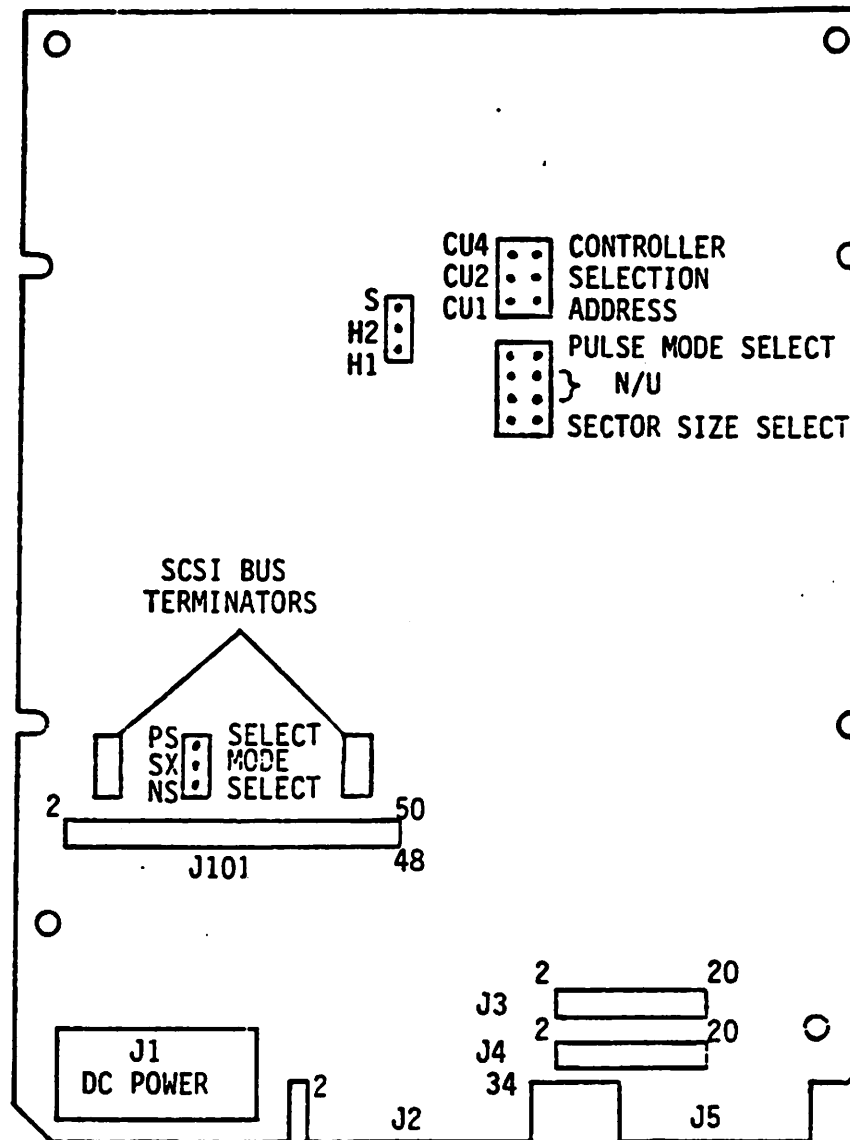


Figure 6.1: 1610-3 Circuit Board Connectors and Option jumpers Locations.

6.2 Physical Specifications.

Width:	5.60 in.	(14.23 cm)
length:	7.75 in.	(19.68 cm)
Height:	0.75 in.	(1.90 cm)
Weight:	8.0 oz.	(0.23 kg)

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7. DATA FORMAT.

7.1 Cylinder and Head limits.

Maximum number of cylinders supported is 2048.
Maximum number of heads supported is 8.

7.2 Track Format.

The nominal track capacity is 10,416 bytes; the minimum track capacity is 10,270 bytes (based on a 1.4 percent speed variation).

Figure 7.2 shows the 1610-3 track format. Note that the size of Gap 4 is dependent on the sector size chosen.

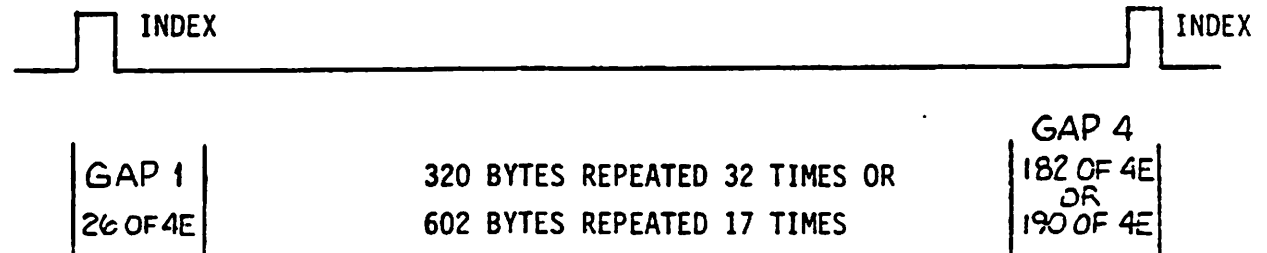


Figure 7.2: Track Format.

7.3 Sector Format.

Figure 7.3.1 shows the sector format. Figure 7.3.2 shows the format of the ID field.



Figure 7.3.1: Sector Format.

```

*--+--+--+--+--+*--+--+--+--+--+*--+--+--+--+--+*
*   C Y L   *X! CYL !Y!  HD *   SECTOR   *
*--+--+--+--+--+*--+--+--+--+--+*--+--+--+--+--+*
                    0       0 - Good Track.
                    0       1 - Alternate Track.
                    1       0 - Bad Track.
                    1       1 - Track Alternated.
  
```

Figure 7.3.2: ID Field Format.

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7.4 Sector Interleave.

Sector interleaving is to tailor host system data transfer speed to disk rotational speed. It allows the host to map contiguous logical blocks on the drive on a non-adjacent physical order. The host system should try different interleave codes to adjust to the maximum data transfer rate.

Example: Interleave code: 3
Maximum sector number: 16

Physical sector:

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Logical sector:

0 6 12 1 7 13 2 8 14 3 9 15 4 10 16 5 11 17

It would take 3 revolutions to transfer the entire track of data.

Note that interleave of zero or one means that consecutive logical blocks to be formatted in consecutive physical order.

Interleave code must be less than the number of sectors per track.

7.5 Defect handling.

The 1610-3 controller supports track level defect handling.

The Format Bad Track command (Opcode 07h) sets a flag to indicate that a track is defective. The controller reports an error if the track is accessed.

The controller allows the host to assign a track as an alternate to a track with defect(s). This is done through the Alternate Track command (Opcode 0Eh). Subsequent access to the defective track will cause the controller to access data from the alternate track automatically. Any direct access to the alternate track causes the controller to report an error.

7.6 Write Precompensation and Reduce Write Current.

The host can specify the cylinder number to start performing write precompensation and reduce write current through the Set Drive Parameter Command (Opcode 0Ch).

The controller automatically enables the write precomp circuitry and selects reduce write current mode on the drive at the specified cylinders. If the host does not specify these locations, the defaults at cylinder 128 for the reduce write current and cylinder 64 for the write precompensation.

8. CONTROLLER COMMAND DESCRIPTION

8.1 Command Block.

An Input/Output request to the 1610-3 is performed by passing a command descriptor block (CDB) to the controller. The CDB consists of six bytes sent by the host during the Command phase. The first byte of the CDB is the command group and operation code. The second to the fourth bytes specify such information as the logical unit number (LUN), and block starting address. The fifth byte specifies the number of blocks to transfer. The sixth byte is the Control Byte.

Refer to the SCSI specification (ANSI X3T9.2) for a detailed description of the structure of the command descriptor block.

8.1.1 Logical Unit Number.

The 1610-3 controller supports a maximum of two logical units. Therefore, the LUN field in the command can take on the value of 0 or 1 only.

6.1.2 Control Byte Format.

The 1610-3 controller expects a special format for the control byte. The control byte is the last byte of all commands.

Bit 7: Disable Retry.

If this bit is set, the controller will not attempt to retry the command.

If this bit is reset, upon encountering any error the controller will retry the command.

NOTE: No retry will be performed on the Check Track Format command.

Bit 6: If this bit is set, the controller will not retry a read of a sector that contains data ECC error before attempting correction. The error status (correctable or uncorrectable) is always reported.

If this bit is reset, the controller will reread the sector before attempting to perform ECC data correction. If the error burst is correctable the error will not be reported.

Bit 5: This bit is used in format operation only. If this bit is set, the controller will write the sectors with the data from the sector buffer. User may select the data pattern by first issuing a Write Sector Buffer command (0Eh).

Bit 4: This bit is set to indicate that the disk drive has servo information prior to index on each track. During formatting, the controller leaves a gap of 300 microseconds in front of the Index and 40 microseconds following the leading edge of the Index pulse.

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Bit 3-0: These four bits are used in combination to select the step rate for the drive. The following step rates are selectable.

BITS				DESCRIPTION
3	2	1	0	
0	0	0	0	3 msec step (default, non buffered)
0	0	0	1	Reserved
0	0	1	0	Reserved
0	0	1	1	Reserved
0	1	0	0	200 usec buffered step
0	1	0	1	70 usec buffered step
0	1	1	0	40 usec buffered step
0	1	1	1	40 usec buffered step
1	0	0	0	40 usec buffered step
1	0	0	1	Reserved
1	0	1	0	Reserved
1	0	1	1	Reserved
1	1	0	0	Reserved
1	1	0	1	Reserved
1	1	1	0	Reserved
1	1	1	1	Reserved

Selecting a "Reserved" step rate causes the controller to use the 3 msec step rate.

8.2 Group 0 Commands.

Tables 8.2 shows the different commands under Group 0.

TABLE 8.2: List of Group 0 Commands.

OPCODE	COMMAND
(00)	TEST UNIT READY
(01)	REZERO UNIT
(03)	REQUEST SENSE
(04)	FORMAT UNIT
(05)	CHECK TRACK FORMAT
(06)	FORMAT TRACK
(07)	FORMAT BAD TRACK
(08)	READ
(09)	READ VERIFY
(0A)	WRITE
(0B)	SEEK
(0C)	SET DRIVE PARAMETERS
(0D)	READ ECC BURST ERROR LENGTH
(0E)	ASSIGN ALTERNATE TRACK
(0F)	WRITE SECTOR BUFFER
(10)	READ SECTOR BUFFER

8.2.1 Test Unit Ready Command (00)

TABLE 8.2.1.: TEST UNIT READY COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	0	0	0	0
Byte-1	-----	LUN	-----	0	0	0	0	0
Byte-2				(00)				
Byte-3				(00)				
Byte-4				(00)				
Byte-5				(00)				

This command selects and verifies that the LUN specified is powered on and ready. The command returns the zero status if the addressed unit is powered on and ready.

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8.2.2 Rezero Unit Command (01)

TABLE 8.2.2.: REZERO UNIT COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	0	0	0	1
Byte-1	-----	LUN	-----	0	0	0	0	0
Byte-2				(00)				
Byte-3				(00)				
Byte-4				(00)				
Byte-5	(*)	0	0	0	0	0	0	0

(*) See description of Control Byte in section 8.1.

Rezero Unit command recalibrates the unit to track zero (00).

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8.2.3 Request Sense Command (03)

TABLE 8.2.3.1.: REQUEST SENSE COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	0	0	1	1
Byte-1	-----	LUN	-----	0	0	0	0	0
Byte-2				(00)				
Byte-3				(00)				
Byte-4				(00)				
Byte-5				(00)				

The REQUEST SENSE command returns four sense bytes. The sense data will be valid for the Check Condition status just presented to the host. Sense data is preserved by the controller for any check condition status, except immediately after a reset.

A REQUEST SENSE command issued with no Check Condition status logged in will return a zero value in Sense Byte 00.

Check Condition status is never returned for this command.

Following an error indication of the status byte with Check Condition, the host may send a REQUEST SENSE command to obtain more detailed information. The format of the sense bytes returned by the controller is shown in Table 8.2.3.2. For a detailed description of the ADDRESS VALID, CLASS and ERROR CODE, see section 8.5.

TABLE 8.2.3.2.: SENSE BYTES FORMAT

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	AD-VALID	---	CLASS (0-3)---		-----	ERROR CODE-----		
Byte-1	-----	LUN	-----		----	LOGICAL BLOCK ADDR (MSB)----		
Byte-2	-----				-----	LOGICAL BLOCK ADDR-----		
Byte-3	-----				-----	LOGICAL BLOCK ADDR (LSB)-----		

If Check Condition status is returned after any format command or the CHECK TRACK FORMAT command, the sense data contains the logical block address of the track where the error occurred. If no error occurred and the host issues a REQUEST SENSE command, then the logical block address of the next track is returned in the sense data.

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8.2.4 Format Unit Command (04)
TABLE 8.2.4.: FORMAT UNIT COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	0	1	0	0
Byte-1	-----	LUN	-----	0	0	0	0	0
Byte-2				(00)				
Byte-3				(00)				
Byte-4	-----	-----	-----	INTERLEAVE	CODE	-----	-----	-----
Byte-5	(*)	0	(*)	(*)	(*)	(*)	(*)	(*)

(*) See description of the Control Byte in Section 8.1.

The FORMAT UNIT command formats the entire media and ensures that all data blocks can be accessed. Before issuing a FORMAT UNIT command, the host should send the SET DRIVE PARAMETER command to specify the drive characteristics. If the drive parameters are not received, the controller attempts to format the drive with default parameters. Refer to the SET DRIVE PARAMETER command description.

8.2.5 Check Track Format Command (05)
TABLE 8.2.5.: CHECK TRACK FORMAT COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	0	1	0	1
Byte-1	-----	LUN	-----	-----	LOGICAL BLOCK ADDR (MSB)	---		
Byte-2	-----	-----	-----	-----	LOGICAL BLOCK ADDRESS	-----		
Byte-3	-----	-----	-----	-----	LOGICAL BLOCK ADDRESS (LSB)	-----		
Byte-4	-----	-----	-----	INTERLEAVE	CODE	-----	-----	-----
Byte-5	(*)	0	0	0	(*)	(*)	(*)	(*)

(*) See description of the Control Byte in Section 8.1.

This command checks the format of the entire track for correctness of all sector ID fields and interleave computation for the track specified by the logical block address. The command does not read the data fields and does not transfer data to the host. The track to be checked is specified by the logical block address of any sector within the track.

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8.2.6 Format Track Command (05)

TABLE 8.2.6.: FORMAT TRACK COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	0	1	1	0
Byte-1	-----	LUN	-----	-----	LOGICAL BLOCK ADDR (MSB)---	---	---	---
Byte-2	-----	-----	-----	-----	LOGICAL BLOCK ADDRESS-----	-----	-----	-----
Byte-3	-----	-----	-----	-----	LOGICAL BLOCK ADDRESS (LSB)-----	-----	-----	-----
Byte-4	-----	-----	-----	-----	INTERLEAVE CODE-----	-----	-----	-----
Byte-5	(*)	0	(*)	(*)	(*)	(*)	(*)	(*)

(*) See description of the Control Byte in Section 8.1.

The FORMAT TRACK command formats all sectors in the track specified by the logical block address of any sector within the track. This operation is similar to the FORMAT UNIT command but limited to one track. User is to ensure that the interleave value set in the CDB is identical to the value previously set for other tracks of the unit.

8.2.7 Format Bad Track (07)

TABLE 8.2.7.: FORMAT BAD TRACK COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	0	1	1	1
Byte-1	-----	LUN	-----	-----	LOGICAL BLOCK ADDR (MSB)---	---	---	---
Byte-2	-----	-----	-----	-----	LOGICAL BLOCK ADDRESS-----	-----	-----	-----
Byte-3	-----	-----	-----	-----	LOGICAL BLOCK ADDRESS (LSB)-----	-----	-----	-----
Byte-4	-----	-----	-----	-----	INTERLEAVE CODE-----	-----	-----	-----
Byte-5	(*)	0	0	(*)	(*)	(*)	(*)	(*)

(*) See description of the Control Byte in Section 8.1.

This command formats all sectors of the track specified by the logical block address of any sector within the track as bad sectors.

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8.2.8 Read Command (08)

TABLE 8.2.8.: READ COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	1	0	0	0
Byte-1	-----	LUN	-----	---	LOGICAL BLOCK ADDR (MSB)	---		
Byte-2	-----	-----	---	LOGICAL BLOCK ADDRESS	-----			
Byte-3	-----	-----	---	LOGICAL BLOCK ADDRESS (LSB)	-----			
Byte-4	-----	-----	---	TRANSFER LENGTH	-----			
Byte-5	(*)	(*)	0	0	(*)	(*)	(*)	(*)

(*) See description of Control Byte in section 8.1.

The READ command transfers the specified number of blocks to the host beginning at the designated logical starting block address. This command is limited to 21 bits of addressing and 256 blocks of transfer. A value of zero in byte 04 means 256 blocks transfer.

If an unrecoverable error occurs during a multiple sector transfer, the controller terminates the transfer at the sector where the error is encountered.

The controller also terminates a multiple sector transfer if a correctable data error occurs and the NO-RETRY bit (bit-7) of the Control Byte is set. An error status sent to the host.

No data is sent to the host if any error occurred.

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8.2.9 Read Verify Command (09)

TABLE 8.2.9.: READ VERIFY COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	1	0	0	1
Byte-1	-----	LUN	-----	---	LOGICAL BLOCK ADDR (MSB)---	---	---	---
Byte-2	-----	-----	-----	-----	-----	-----	-----	-----
Byte-3	-----	-----	-----	-----	-----	-----	-----	-----
Byte-4	-----	-----	-----	-----	-----	-----	-----	-----
Byte-5	(*)	(*)	0	0	(*)	(*)	(*)	(*)

(*) See description of Control Byte in section 8.1.

This command functions the same as the Read Command, except that no data is sent to the host. This command may be used to verify the data following a write or format operation.

8.2.10 Write Command (0A)

Table 8.2.10.: WRITE COMMAND.

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	1	0	1	0
Byte-1	-----	LUN	-----	---	LOGICAL BLOCK ADDR (MSB)---	---	---	---
Byte-2	-----	-----	-----	-----	-----	-----	-----	-----
Byte-3	-----	-----	-----	-----	-----	-----	-----	-----
Byte-4	-----	-----	-----	-----	-----	-----	-----	-----
Byte-5	(*)	0	0	0	(*)	(*)	(*)	(*)

(*) See description of Control Byte in section 8.1.

The WRITE command transfers the specified number of blocks beginning at the designated logical starting block address. This command is limited to 21 bit addressing and 256 blocks of transfer. A value of zero in byte 04 means 256 bytes transfer.

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8.2.11 Seek Command (0B)

Table 8.2.11: SEEK COMMAND.

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	1	0	1	1
Byte-1	-----	LUN	-----	---	LOGICAL BLOCK ADDR (MSB)	---		
Byte-2	-----	---	LOGICAL BLOCK ADDRESS	-----				
Byte-3	-----	---	LOGICAL BLOCK ADDRESS (LSB)	-----				
Byte-4				(00)				
Byte-5	(*)	0	0	0	(*)	(*)	(*)	(*)

(*) See description of Control Byte in section 8.1.

The SEEK command requests the unit to prepare itself to transfer data from the specified address in minimal time. As READ and WRITE commands are performed with implied seek, it is not mandatory to use this command. The command is limited to 21 bit addressing.

If Buffered Step mode is specified in the Control Byte, the controller will issue the seek step pulses to the disk drive and report command complete to the host before receiving SEEK COMPLETE status from the drive. This allows the user to perform overlap seek operations. If another command is issued by the host for the drive unit that is still seeking, then the controller will wait until the seek is completed before executing the new command.

The TEST UNIT READY command can be used during Buffer Step mode to determine if a seek request is completed. A Check Condition status is returned if the drive is still seeking. The returned sense data is DRIVE STILL SEEKING (Class:0 Code:8).

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8.2.12 Set Drive Parameters Command (OC).

TABLE 8.2.12.1: SET DRIVE PARAMETER COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	1	1	0	0
Byte-1				(00)				
Byte-2				(00)				
Byte-3				(00)				
Byte-4				(00)				
Byte-5				(00)				

The host specifies the disk parameters of the drives connected to the controller by sending the SET DRIVE PARAMETER command. This should be done after power-on or a RESET to the controller. The drive parameters are sent to the controller during the DATA OUT phase of the command execution. The parameters are stored in the controller RAM.

The parameters and characteristics of both drives connected to the controller are initialized when this command is issued. Therefore, if drives with different parameters are used, this command must be sent to the controller each time when switching between drive.

The format of the drive parameter block is described in Table 8.2.12.2.

Table 8.2.12.2: Drive Parameter Format.

Byte-0:	high (615)	2	Maximum number of Cylinders (MSB).
Byte-1:	low (615)	103	Maximum number of Cylinders (LSB).
Byte-2:	4		Maximum number of Heads.
Byte-3:	high (615)	2	Reduce Write Current Cylinder (MSB).
Byte-4:	low (615)	103	Reduce Write Current Cylinder (LSB).
Byte-5:	high (122)	0	Write Precompensation Cylinder (MSB).
Byte-6:	low (122)	128	Write Precompensation Cylinder (LSB).
Byte-7:	8		Maximum ECC data burst length

stop option 7
The maximum number of cylinders and heads supported by the 1610-3 controller is defined in Section 7.1 of this document.

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Upon reset or power-up, the controller defaults to the drive parameters as shown in table 8.2.12.3.

Table 8.2.12.3: Default Drive Parameters.

HEX VALUE		
Byte-0:	00	
Byte-1:	99	Maximum number of Cylinders = 153.
Byte-2:	04	Maximum number of Heads = 4.
Byte-3:	00	
Byte-4:	80	Reduce Writer Current Cylinder = 128.
Byte-5:	00	
Byte-6:	40	Write Precompensation Cylinder = 64.
Byte-7:	08	Maximum ECC data burst length correctable = 8 bits. (Any larger value defaults to 8.)

8.2.13 Read ECC Burst Error Length Command (0D).

TABLE 8.2.13.1: READ ECC BURST ERROR LENGTH COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	1	1	0	1
Byte-1				(00)				
Byte-2				(00)				
Byte-3				(00)				
Byte-4				(00)				
Byte-5				(00)				

This command transfers one byte of information to the host indicating the ECC burst error length of an error just encountered. This command is only valid immediately following a Correctable Data Error code (18h).

8.2.14 Assign Alternate Track Command (OE)

TABLE 8.2.14.1: ASSIGN ALTERNATE TRACK COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	0	1	1	1
Byte-1	----- LUN -----			----- LOGICAL BLOCK ADDR (MSB) -----				
Byte-2	----- LOGICAL BLOCK ADDRESS -----							
Byte-3	----- LOGICAL BLOCK ADDRESS (LSB) -----							
Byte-4	----- INTERLEAVE CODE -----							
Byte-5	(*)	0	(*)	(*)	(*)	(*)	(*)	(*)

(*) See description of the Control Byte in Section 8.1.

The ASSIGN ALTERNATE TRACK command allocates an alternate track (whose address is specified in a 3 byte field sent in the DATA OUT phase of the command) to the track specified by the logical block address in the CDB. The controller modifies the format of both the alternated and the alternate track by changing the ID fields of all sectors of those tracks.

Table 8.2.14.2 shows the format of the 3 byte field that is sent during the DATA OUT phase to specify the alternate track address

TABLE 8.2.14.2: ALTERNATE TRACK ADDRESS

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	----LOGICAL BLOCK ADDR (MSB)---				
Byte-1	-----LOGICAL BLOCK ADDRESS-----							
Byte-2	-----LOGICAL BLOCK ADDRESS (LSB)-----							

Attempts to access the logical block(s) of a track that is alternated causes the controller to access the blocks at the alternate track. Attempts to access any sector in the alternate track directly causes a CHECK CONDITION status.

Alternate tracks cannot be allocated as alternated tracks to other alternate tracks. Only one level of assignment is allowed. A track cannot be alternated to itself.

8.2.15 Write Sector Buffer Command (3F).

TABLE 8.2.15: WRITE SECTOR BUFFER COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	0	1	1	1	1
Byte-1				(00)				
Byte-2				(00)				
Byte-3				(00)				
Byte-4				(00)				
Byte-5				(00)				

This command allows the user to send to the controller RAM buffer a sector of data.

After issuing this command, subsequent format operation may write disk with the data in the buffer, if bit 5 in the control byte of the format command block is set.

This command may also be used to test the RAM memory by writing a test pattern and reading it back with the Read Sector Buffer command.

8.2.16 Read Sector Buffer Command (10).

TABLE 8.2.16.1: READ SECTOR BUFFER COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	0	0	0	1	0	0	0	0
Byte-1				(00)				
Byte-2				(00)				
Byte-3				(00)				
Byte-4				(00)				
Byte-5				(00)				

The controller will not send the data to the host if an uncorrectable data error is detected. Therefore, this command may be used to retrieve the data in the controller's buffer.

This command is also used to test the RAM buffer.

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8.3 Group 7 Commands.

Tables 8.3 shows the Group 7 commands supported by the 1610-3 controller.

TABLE 8.3: List of Group 7 Commands.

OPCODE	COMMAND
(E0)	RAM DIAGNOSTIC
(E3)	DRIVE DIAGNOSTIC
(E4)	CONTROLLER DIAGNOSTIC
(E7)	REQUEST LOGOUT

8.3.1 RAM Diagnostic Command (E0).

TABLE 8.3.1.: RAM DIAGNOSTIC COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	1	1	1	0	0	0	0	0
Byte-1				(00)				
Byte-2				(00)				
Byte-3				(00)				
Byte-4				(00)				
Byte-5				(00)				

This command performs a data pattern test on the internal controller 1K RAM. The controller automatically performs this function at power-up.

8.3.2 Drive Diagnostic Command (E3).

TABLE 8.3.2.: DRIVE DIAGNOSTIC COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	1	1	1	0	0	0	1	1
Byte-1				(00)				
Byte-2				(00)				
Byte-3				(00)				
Byte-4				(00)				
Byte-5				(00)				

This diagnostic command reads sector zero on all tracks sequentially.

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8.3.3 Controller Diagnostic Command (E4).

TABLE 8.3.3.: CONTROLLER DIAGNOSTIC COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	1	1	1	0	0	1	0	0
Byte-1				(00)				
Byte-2				(00)				
Byte-3				(00)				
Byte-4				(00)				
Byte-5				(00)				

This command causes the controller to perform an internal diagnostic. The controller first verifies the checksum of the EPROM. The ECC circuitry is then tested by shifting in a data pattern and verifying the ECC pattern generated.

8.3.4 Request Logout Command (E7).

TABLE 8.3.4.1: REQUEST LOGOUT COMMAND

	Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
Byte-0	1	1	1	0	0	1	1	1
Byte-1	-----LUN-----			0	0	0	0	0
Byte-2				(00)				
Byte-3				(00)				
Byte-4				(00)				
Byte-5				(00)				

The controller keeps track of the error statistics for each connected drive. This command causes the controller to return the 8 bytes of error statistics associated with the specific LUN during the DATA IN phase. The format of the error statistics returned is shown in Table 8.3.4.2.

The error log area is cleared after completing this command.

Table 8.3.4.2: Error Logout Data Format.

Byte-0:	Non-recoverable Error Count (MSB).
Byte-1:	Non-recoverable Error Count (LSB).
Byte-2:	Recoverable Errors (MSB).
Byte-3:	Recoverable Errors (LSB).
Byte-4:	Soft ECC Errors (MSB).
Byte-5:	Soft ECC Errors (LSB).
Byte-6:	Correctable ECC Errors (MSB).
Byte-7:	Correctable ECC Errors (LSB).

- Non-recoverable errors are errors that cannot be overcome after all retries, except unrecoverable ECC errors.
- Recoverable errors are ones that are overcome after one or more retries. This count excludes the recoverable ECC errors.
- Soft ECC errors are ECC errors that are detected on the first read operation which are then read with no errors during retry operations.
- Correctable ECC errors are errors that are correctable using the ECC correction scheme.

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8.4 Completion Status Byte.

When the 1610-3 controller completes a command, a byte of status is always sent to the host during the STATUS phase. This is used to indicate to the host if the command is completed successfully.

The format of the status byte is shown in Table 8.4.

Table 8.4.: Status Byte Format.

Bit-7	Bit-6	Bit-5	Bit-4	Bit-3	Bit-2	Bit-1	Bit-0
<-----LUN----->			0	0	0	ERROR	0

- Bit-1: ERROR; This bit is set if an error occurred during command execution. The host may perform a REQUEST SENSE command (03) to obtain more detailed information.

8.5 Sense Bytes Format.

The REQUEST SENSE command (03) returns a block of four bytes of sense information to the host. See Section 8.2.3.

The first byte of the sense data contains the following fields:

- Bit-7: ADDRESS VALID; If set, then bytes 01 through 03 of the sense data represent the address at which the error occurred.
- Bit-4 to 6: CLASS; Specifies the error class.
- Bit-0 to 3: ERROR CODE; Specifies the type of error.

8.5.1 Class 0: Drive Error Codes.

HEX VALUE

(0)	No Sense.
(1)	No Index Signal From Drive Selected.
(2)	No Seek Complete.
(3)	Drive Fault.
(4)	Drive Not Ready.
(5)	Not Used.
(6)	No Track 00.
(7)	Not Used.
(8)	Seek In Progress.
(9 - F)	Not Used.

8.5.2 Class 1: Controller Error Codes.

HEX VALUE

(0)	ID CRC Error.
(1)	Uncorrectable Data Error.
(2)	ID Address Mark Not Found.
(3)	Not Used.
(4)	Record Not Found.
(5)	Seek Error.
(8)	Correctable Data Check.
(9)	Bad Block Found.
(A)	Format Error.
(B)	Not Used.
(C)	Direct Access to an Alternate track.
(D)	Alternate track already assigned.
(E)	Assigned track of an Alternated track not formatted as an Alternate track.
(F)	The Alternate track is assigned to itself on an Assign Alternate Track command.

8.5.3 Class 2: Command Error Codes.

(0)	Invalid Command.
(1)	Illegal Block Address.
(2)	Illegal Function for Current Drive Type.
(3 - F)	Not Used.

8.5.4 Class 3: Miscellaneous Error Codes.

(0)	RAM error, Detected During Diagnostic.
(1)	EPROM Checksum error.
(2)	ECC diagnostic error.
(3 - F)	Not Used.

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9.1 Environmental Specifications

	Storage	Transit	Non-Oper.	Operating
Temp.	-40 to 140oF (-40 to 60oC)	-40 to 140oF (-40 to 60oC)	50 to 115oF (10 to 46oC)	50 to 115oF (10 to 46oC)
Temp. Gradient	27oF/hr (15oC/hr)	27oF/hr (15oC/hr)	18oF/hr (10oC/hr)	18oF/hr (10oC/hr)
Relative Humidity	10 - 95% No Conden- sation.	10 - 95% No Conden- sation.	10 - 80% No Conden- sation.	10 - 80% No Conden- sation.
Humidity Gradient	10%/hr	10%/hr	10%/hr	10%/hr
Altitude	0-40,000ft. (0-12,192 m)	0-40,000ft. (0-12,192 m)	0-40,000ft. (0-12,192 m)	0-10,000ft (0-3,048 m)

The controller normally does not require fans in standard operating environments where airflow is not restricted.

9.2 Power Requirements.

J1 is a four pin connector which supplies DC power to the controller. Table 8.2 shows the pin assignment for this connector.

Table 8.2: J1 Connector Power Requirements

Pin	Voltage	Current Nom	Current Max	Ripple
1	+12VDC (+/-5%)	40 ma	50 ma	100 mv
2	+12V RETURN			
3	+ 5V RETURN			

9.3 Compatibility

The 1610-3 controller is designed to be compatible with the Xebec S1410A which runs with the standard firmware set (Xebec P/N 104788). Due to hardware limitations and differences, the 1610-3 controller differs from the Xebec S1410A as follows:

1610-3	S1410A with standard firmware
- Commands not supported: Diagnostic Read Long and Diagnostic Write Long.	N/A
- Number of heads supported: 8 Heads	16 Heads
- Supported step rates: 3 mSec, 200 uSec, 70 uSec, and 40 uSec.	3 mSec, 200 uSec, 70 uSec, 30 uSec, 15 uSec and 12 uSec.
- Write precompensation values: 0 or +/- 12 nSec.	Determined by bit P (MSB of write precomp. parameters) as follows: P = 0: 0 or +/- 5 nSec P = 1: 5 or +/- 15 nSec.
- Selection phase timing: Minimum time for assertion of SEL signal: . 50 uSec: normal mode 100 nSec . 100 nSec: pulse mode (limited to 1 controller per SCSI bus).	
BSY responses to SEL: 50 uSec	50 nSec.
- ECC error burst length: 8 bits Max.	11 bits Max. (1000 times higher probability of miscorrection).
- Power requirements: + 5.0 VDC at 1.7 A Max. +12.0 VDC at .05 A Max.	+ 5.0 VDC at 1.3 A Max. +12.0 VDC at .006 A Max.

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL CAPTION VALUES ARE IN MICROHMS. $1\mu = 100\text{ NS}$.
2. ALL RESISTION VALUES ARE IN OHMS. $1\text{K} = 1000\Omega$.
3. ALL DIODES ARE 1N4148.
4. $\square$ INDICATES CUT-SPACE OPTION.
5. $\square$ INDICATES JUMPER OPTION.
6. $\square$ INDICATES ADD WIRE OPTION.
7. $\square$ INDICATES OFF-SHEET CONNECTION.
8. $\square$ NOT INSTALLED

U10	U11	U12	U13	U14	U15	U16	U17	U18	U19	U20	U21	U22	U23	U24	U25	U26	U27	U28	U29	U30	U31	U32	U33	U34	U35	U36	U37	U38	U39	U40	U41	U42	U43	U44	U45	U46	U47	U48	U49	U50	U51	U52	U53	U54	U55	U56	U57	U58	U59	U60	U61	U62	U63	U64	U65	U66	U67	U68	U69	U70	U71	U72	U73	U74	U75	U76	U77	U78	U79	U80	U81	U82	U83	U84	U85	U86	U87	U88	U89	U90	U91	U92	U93	U94	U95	U96	U97	U98	U99	U100	U101	U102	U103	U104	U105	U106	U107	U108	U109	U110	U111	U112	U113	U114	U115	U116	U117	U118	U119	U120	U121	U122	U123	U124	U125	U126	U127	U128	U129	U130	U131	U132	U133	U134	U135	U136	U137	U138	U139	U140	U141	U142	U143	U144	U145	U146	U147	U148	U149	U150	U151	U152	U153	U154	U155	U156	U157	U158	U159	U160	U161	U162	U163	U164	U165	U166	U167	U168	U169	U170	U171	U172	U173	U174	U175	U176	U177	U178	U179	U180	U181	U182	U183	U184	U185	U186	U187	U188	U189	U190	U191	U192	U193	U194	U195	U196	U197	U198	U199	U200	U201	U202	U203	U204	U205	U206	U207	U208	U209	U210	U211	U212	U213	U214	U215	U216	U217	U218	U219	U220	U221	U222	U223	U224	U225	U226	U227	U228	U229	U230	U231	U232	U233	U234	U235	U236	U237	U238	U239	U240	U241	U242	U243	U244	U245	U246	U247	U248	U249	U250	U251	U252	U253	U254	U255	U256	U257	U258	U259	U260	U261	U262	U263	U264	U265	U266	U267	U268	U269	U270	U271	U272	U273	U274	U275	U276	U277	U278	U279	U280	U281	U282	U283	U284	U285	U286	U287	U288	U289	U290	U291	U292	U293	U294	U295	U296	U297	U298	U299	U300	U301	U302	U303	U304	U305	U306	U307	U308	U309	U310	U311	U312	U313	U314	U315	U316	U317	U318	U319	U320	U321	U322	U323	U324	U325	U326	U327	U328	U329	U330	U331	U332	U333	U334	U335	U336	U337	U338	U339	U340	U341	U342	U343	U344	U345	U346	U347	U348	U349	U350	U351	U352	U353	U354	U355	U356	U357	U358	U359	U360	U361	U362	U363	U364	U365	U366	U367	U368	U369	U370	U371	U372	U373	U374	U375	U376	U377	U378	U379	U380	U381	U382	U383	U384	U385	U386	U387	U388	U389	U390	U391	U392	U393	U394	U395	U396	U397	U398	U399	U400	U401	U402	U403	U404	U405	U406	U407	U408	U409	U410	U411	U412	U413	U414	U415	U416	U417	U418	U419	U420	U421	U422	U423	U424	U425	U426	U427	U428	U429	U430	U431	U432	U433	U434	U435	U436	U437	U438	U439	U440	U441	U442	U443	U444	U445	U446	U447	U448	U449	U450	U451	U452	U453	U454	U455	U456	U457	U458	U459	U460	U461	U462	U463	U464	U465	U466	U467	U468	U469	U470	U471	U472	U473	U474	U475	U476	U477	U478	U479	U480	U481	U482	U483	U484	U485	U486	U487	U488	U489	U490	U491	U492	U493	U494	U495	U496	U497	U498	U499	U500	U501	U502	U503	U504	U505	U506	U507	U508	U509	U510	U511	U512	U513	U514	U515	U516	U517	U518	U519	U520	U521	U522	U523	U524	U525	U526	U527	U528	U529	U530	U531	U532	U533	U534	U535	U536	U537	U538	U539	U540	U541	U542	U543	U544	U545	U546	U547	U548	U549	U550	U551	U552	U553	U554	U555	U556	U557	U558	U559	U560	U561	U562	U563	U564	U565	U566	U567	U568	U569	U570	U571	U572	U573	U574	U575	U576	U577	U578	U579	U580	U581	U582	U583	U584	U585	U586	U587	U588	U589	U590	U591	U592	U593	U594	U595	U596	U597	U598	U599	U600	U601	U602	U603	U604	U605	U606	U607	U608	U609	U610	U611	U612	U613	U614	U615	U616	U617	U618	U619	U620	U621	U622	U623	U624	U625	U626	U627	U628	U629	U630	U631	U632	U633	U634	U635	U636	U637	U638	U639	U640	U641	U642	U643	U644	U645	U646	U647	U648	U649	U650	U651	U652	U653	U654	U655	U656	U657	U658	U659	U660	U661	U662	U663	U664	U665	U666	U667	U668	U669	U670	U671	U672	U673	U674	U675	U676	U677	U678	U679	U680	U681	U682	U683	U684	U685	U686	U687	U688	U689	U690	U691	U692	U693	U694	U695	U696	U697	U698	U699	U700	U701	U702	U703	U704	U705	U706	U707	U708	U709	U710	U711	U712	U713	U714	U715	U716	U717	U718	U719	U720	U721	U722	U723	U724	U725	U726	U727	U728	U729	U730	U731	U732	U733	U734	U735	U736	U737	U738	U739	U740	U741	U742	U743	U744	U745	U746	U747	U748	U749	U750	U751	U752	U753	U754	U755	U756	U757	U758	U759	U760	U761	U762	U763	U764	U765	U766	U767	U768	U769	U770	U771	U772	U773	U774	U775	U776	U777	U778	U779	U780	U781	U782	U783	U784	U785	U786	U787	U788	U789	U790	U791	U792	U793	U794	U795	U796	U797	U798	U799	U800	U801	U802	U803	U804	U805	U806	U807	U808	U809	U810	U811	U812	U813	U814	U815	U816	U817	U818	U819	U820	U821	U822	U823	U824	U825	U826	U827	U828	U829	U830	U831	U832	U833	U834	U835	U836	U837	U838	U839	U840	U841	U842	U843	U844	U845	U846	U847	U848	U849	U850	U851	U852	U853	U854	U855	U856	U857	U858	U859	U860	U861	U862	U863	U864	U865	U866	U867	U868	U869	U870	U871	U872	U873	U874	U875	U876	U877	U878	U879	U880	U881	U882	U883	U884	U885	U886	U887	U888	U889	U890	U891	U892	U893	U894	U895	U896	U897	U898	U899	U900	U901	U902	U903	U904	U905	U906	U907	U908	U909	U910	U911	U912	U913	U914	U915	U916	U917	U918	U919	U920	U921	U922	U923	U924	U925	U926	U927	U928	U929	U930	U931	U932	U933	U934	U935	U936	U937	U938	U939	U940	U941	U942	U943	U944	U945	U946	U947	U948	U949	U950	U951	U952	U953	U954	U955	U956	U957	U958	U959	U960	U961	U962	U963	U964	U965	U966	U967	U968	U969	U970	U971	U972	U973	U974	U975	U976	U977	U978	U979	U980	U981	U982	U983	U984	U985	U986	U987	U988	U989	U990	U991	U992	U993	U994	U995	U996	U997	U998	U999	U1000	U1001	U1002	U1003	U1004	U1005	U1006	U1007	U1008	U1009	U1010	U1011	U1012	U1013	U1014	U1015	U1016	U1017	U1018	U1019	U1020	U1021	U1022	U1023	U1024	U1025	U1026	U1027	U1028	U1029	U1030	U1031	U1032	U1033	U1034	U1035	U1036	U1037	U1038	U1039	U1040	U1041	U1042	U1043	U1044	U1045	U1046	U1047	U1048	U1049	U1050	U1051	U1052	U1053	U1054	U1055	U1056	U1057	U1058	U1059	U1060	U1061	U1062	U1063	U1064	U1065	U1066	U1067	U1068	U1069	U1070	U1071	U1072	U1073	U1074	U1075	U1076	U1077	U1078	U1079	U1080	U1081	U1082	U1083	U1084	U1085	U1086	U1087	U1088	U1089	U1090	U1091	U1092	U1093	U1094	U1095	U1096	U1097	U1098	U1099	U1100	U1101	U1102	U1103	U1104	U1105	U1106	U1107	U1108	U1109	U1110	U1111	U1112	U1113	U1114	U1115	U1116	U1117	U1118	U1119	U1120	U1121	U1122	U1123	U1124	U1125	U1126	U1127	U1128	U1129	U1130	U1131	U1132	U1133	U1134	U1135	U1136	U1137	U1138	U1139	U1140	U1141	U1142	U1143	U1144	U1145	U1146	U1147	U1148	U1149	U1150	U1151	U1152	U1153	U1154	U1155	U1156	U1157	U1158	U1159	U1160	U1161	U1162	U1163	U1164	U1165	U1166	U1167	U1168	U1169	U1170	U1171	U1172	U1173	U1174	U1175	U1176	U1177	U1178	U1179	U1180	U1181	U1182	U1183	U1184	U1185	U1186	U1187	U1188	U1189	U1190	U1191	U1192	U1193	U1194	U1195	U1196	U1197	U1198	U1199	U1200	U1201	U1202	U1203	U1204	U1205	U1206	U1207	U1208	U1209	U1210	U1211	U1212	U1213	U1214	U1215	U1216	U1217	U1218	U1219	U1220	U1221	U1222	U1223	U1224	U1225	U1226	U1227	U1228	U1229	U1230	U1231	U1232	U1233	U1234	U1235	U1236	U1237	U1238	U1239	U1240	U1241	U1242	U1243	U1244	U1245	U1246	U1247	U1248	U1249	U1250	U1251	U1252	U1253	U1254	U1255	U1256	U1257	U1258	U1259	U1260	U1261	U1262	U1263	U1264	U1265	U1266	U1267	U1268	U1269	U1270	U1271	U1272	U1273	U1274	U1275	U1276	U1277	U1278	U1279	U1280	U1281	U1282	U1283	U1284	U1285	U1286	U1287	U1288	U1289	U1290	U1291	U1292	U1293	U1294	U1295	U1296	U1297	U1298	U1299	U1300	U1301	U1302	U1303	U1304	U1305	U1306	U1307	U1308	U1309	U1310	U1311	U1312	U1313	U1314	U1315	U1316	U1317	U1318	U1319	U1320	U1321	U1322	U1323	U1324	U1325	U1326	U1327	U1328	U1329	U1330	U1331	U1332	U1333	U1334	U1335	U1332
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12 WIRE FOR REV. 5 ON 12/70

